LESSON 1

Functions

Exercise 1.1

1. Tick (✓) the correct option for the given questions:

(A) Which of the following statements is incorrect regarding the composite functions?

a) If $f : A \rightarrow B$ and $g : B \rightarrow C$ exist, and $A \rightarrow C$ is also a function, it is composite.
b) If $g : A \rightarrow B$ and $f : B \rightarrow C$ are one-to-one and onto, then $A \rightarrow C$ is composite.
c) If $g : A \rightarrow B$ and $f : B \rightarrow C$ are defined, the function $A \rightarrow C$ is denoted by $g \circ f$.
d) If $f : A \rightarrow B$ and $g : B \rightarrow C$ are defined, the relation $C \rightarrow A$ cannot be composite.

Ans: c) If $g : A \rightarrow B$ and $f : B \rightarrow C$ are defined, the function $A \rightarrow C$ is denoted by $g \circ f$.

(B) In the given mapping diagram, how is the function defined from A to C denoted?

a) $fg(x)$ **b)** $gf(x)$
c) $ff(x)$ **d)** $gg(x)$

Ans: a) $fg(x)$

(C) If $f(x) = \{(3, -1), (2, 0), (1, 1)\}$ and $g(x) = \{(-1, 4), (0, 5), (1, 6)\}$, what is $gf(x)$?

a) $\{(4, 3), (2, 5), (1, 6)\}$ **b)** $\{(3, 4), (2, 5), (2, 6)\}$
c) $\{(3, 4), (1, 5), (2, 6)\}$ **d)** $\{(3, 4), (2, 5), (1, 6)\}$

Ans: d) $\{(3, 4), (2, 5), (1, 6)\}$

Solution:

Here, $gf(x) = g(f(x))$
$gf(3) = g(f(3)) = g(-1) = 4$
$gf(2) = g(f(2)) = g(0) = 5$
$gf(1) = g(f(1)) = g(1) = 6$
$\therefore\ gf(x) = \{(3, 4), (2, 5), (1, 6)\}$

(D) If $f(x) = 3x - 1$ and $g(x) = x + 1$, what is $fg(x)$?

a) $3x - 1$ **b)** $3x + 1$
c) $3x - 2$ **d)** $3x + 2$

Ans: d) $3x + 2$

Solution:

$fg(x) = f(g(x)) = f(x + 1) = 3(x + 1) - 1 = 3x + 3 - 1 = 3x + 2$

(E) If $f(x) = 1 - 3x$ and $g(x) = 2x + 1$, what is $gf(\frac{1}{2})$?

a) 3 **b)** 1 **c)** 2 **d)** 0

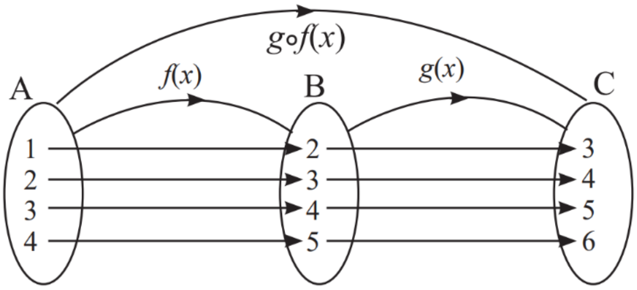
Ans: d) 0

Solution:

$f(½) = 1 - 3(½) = 1 - 3/2 = -½$
$gf(½) = g(f(½)) = g(-½) = 2(-½) + 1 = -1 + 1 = 0$

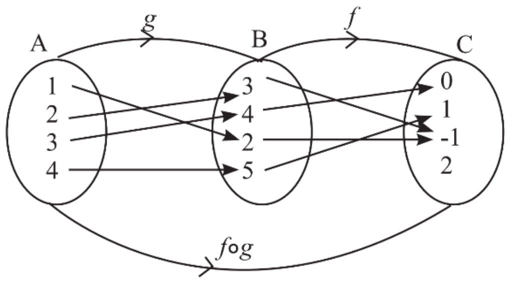
2. Define composite function with a mapping diagram.

If functions f: A→B and g: B→C are defined, and a relation A→C is also defined as a function, then the function A→C is called the composite function of f and g. This is denoted in the form $g∘f$: A→C or $g∘f(x)$ or $g∘f$ or $gf(x)$ or gf.



---

3. In the given mapping diagram, functions g : A → B and f : B → C are defined. Answer the following:



(a) Write function g : A → B as a set of ordered pairs.

Ans: Function g:A→B = {(1, 2), (2, 3), (3, 4), (4, 5)}

(b) Write function f : B → C as a set of ordered pairs.

Ans: Function f:B→C = {(3, -1), (4, 0), (2, -1), (5, 1)}

(c) Find the values of $f∘g(1)$, $f∘g(2)$, $f∘g(3)$, and $f∘g(4)$.

Ans: $f∘g(1) = f(g(1)) = f(2) = -1$

$f∘g(2) = f(g(2)) = f(3) = -1$

$f∘g(3) = f(g(3)) = f(4) = 0$

$f∘g(4) = f(g(4)) = f(5) = 1$

(d) Write the composite function $f∘g$ as a set of ordered pairs.

Ans: By combining the starting input from set A with the final resulting output in set C:

Composite function $f \circ g = \{(1,-1),(2,-1),(3,0),(4,1)\}$

---

4. If $f(x) = \{(1, 2), (2, 3), (3, 4)\}$ and $g(x) = \{(2, 4), (3, 5), (4, 6)\}$, show $gf(x)$ in a mapping diagram and write it as a set of ordered pairs.

Solution:

Given, $f(x) = \{(1,2),(2,3),(3,4)\}$

$g(x) = \{(2,4),(3,5),(4,6)\}$

then,

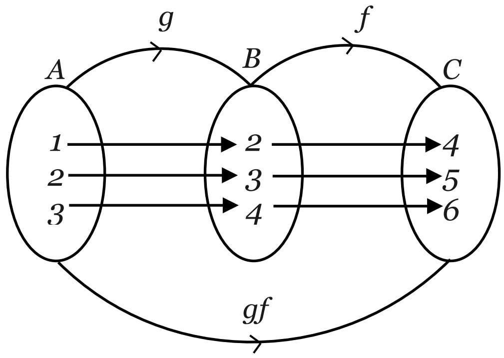
for $x=1$: $gf(1) = g(f(1)) = g(2) = 4$

for $x=2$: $gf(2) = g(f(2)) = g(3) = 5$

for $x=3$: $gf(3) = g(f(3)) = g(4) = 6$

Ans: $gf(x) = \{(1, 4), (2, 5), (3, 6)\}$

Mapping Diagram :



5. If $f(x) = 3x + 2$ and $g(x) = 5 - 3x$, find the value of the following composite functions:

a. $gf(x)$

Solution:

Given, $f(x) = 3x+2$

$g(x) = 5-3x$

then,

$$gf(x) = g(f(x)) = g(3x + 2)$$
$$= 5 - 3(3x + 2)$$
$$= 5 - 9x - 6$$
$$= -9x - 1$$

Ans: $gf(x) = -9x - 1$

b. $fg(x)$

Solution:

$$fg(x) = f(g(x)) = f(5 - 3x)$$
$$= 3(5 - 3x) + 2$$
$$= 15 - 9x + 2$$
$$= 17 - 9x$$

Ans: $fg(x) = 17 - 9x$

c. $ff(x)$

Solution:

$ff(x) = f(f(x)) = f(3x + 2)$
$= 3(3x + 2) + 2$
$= 9x + 6 + 2$
$= 9x + 8$

Ans: $ff(x) = 9x + 8$

d. $ff(3)$

Solution:

$ff(x) = 9x + 8$ (from question c)
$ff(3) = 9(3) + 8 = 27 + 8 = 35$

Ans: $ff(3) = 35$

---

6. If $f(x) = \frac{2x-1}{x+3}$ $(x \neq -3)$ and $g(x) = \frac{3x+2}{x-4}$ $(x \neq 4)$, find the value of $fg(x)$.

Solution:

Here, Given,

$$f(x) = \frac{2x-1}{x+3}, \quad g(x) = \frac{3x+2}{x-4}$$

Now,

$$fg(x) = f(g(x))$$
$$= f\left(\frac{3x+2}{x-4}\right)$$
$$= \frac{2\left(\frac{3x+2}{x-4}\right)-1}{\left(\frac{3x+2}{x-4}\right)+3}$$
$$= \frac{\frac{2(3x+2)-1(x-4)}{x-4}}{\frac{(3x+2)+3(x-4)}{x-4}}$$
$$= \frac{2(3x+2)-(x-4)}{(3x+2)+3(x-4)}$$
$$= \frac{6x+4-x+4}{3x+2+3x-12}$$
$$= \frac{5x+8}{6x-10}$$

Ans: $\therefore fg(x) = \frac{5x+8}{6x-10}$

---

7. If $f(x) = 2x + 5$ and $g(x) = \frac{x-5}{2}$, prove that $fg(x)$ is an identity function.

Solution:

Here, Given,

$$f(x) = 2x + 5$$
$$g(x) = \frac{x-5}{2}$$

Now,

$$fg(x) = f(g(x))$$
$$= f\left(\frac{x-5}{2}\right)$$
$$= 2\left(\frac{x-5}{2}\right) + 5$$
$$= (x-5) + 5$$
$$= x - 5 + 5$$
$$= x$$

Since $fg(x) = x$, the output of the function is identical to its input.

Ans: $\therefore fg(x)$ is an identity function.

---

8. a. If $f(x) = \dfrac{x-3}{2}$ and $fg(x) = x$, find the value of $g(x)$.

Solution:

Here,Given,

$f(x) = \frac{x-3}{2}$ $\quad and \quad$ $fg(x) = x$

Now,

$f(g(x)) = x$

Or, $\frac{g(x)-3}{2} = x$

Or, $g(x) - 3 = 2x$

Or, $g(x) = 2x + 3$

Ans: $\therefore g(x) = 2x + 3$

---

8. b. If $g(x) = 4 - x$ and $gh(x) = 11 - 2x$, find the value of $h(x)$.

Solution:

Here,

$g(x) = 4 - x$ $\quad and \quad$ $gh(x) = 11 - 2x$

Now,

$g(h(x)) = 11 - 2x$

Or, $4 - h(x) = 11 - 2x$

Or, $4 - 11 + 2x = h(x)$

Or, $2x - 7 = h(x)$

Ans: $\therefore h(x) = 2x - 7$

---

9. a. If $f(x) = \frac{6}{x-2}$ $(x \neq 2)$, $g(x) = ax^2 - 1$ and $gf(5) = 7$, find the value of a.

Solution:

Here,Given,

$f(x) = \frac{6}{x-2}$

$g(x) = ax^2 - 1$

$gf(5) = 7$

a = ?

Now,

$gf(5) = g(f(5))$

Or, $gf(5) = g(\frac{6}{5-2})$

Or, $gf(5) = g(\frac{6}{3}) = g(2)$

Or, $gf(5) = g(2) = 7$

now,

$g(2) = 7$

Or, $a(2)^2 - 1 = 7$

Or, $4a = 8$

Or, a = 2

Ans: ∴ a = 2

---

9. b. If $f(x) = ax + 5$, $g(x) = 8x + 13$ and $gf(5) = 93$, find the value of a.

Solution:

Here,

$f(x) = ax + 5$

$g(x) = 8x + 13$

$gf(5) = 93$

Now,

$gf(5) = g(5a + 5)$

Or, $93 = g(5a + 5)$

Or, $93 = 8(5a + 5) + 13$

Or, $93 = 40a + 40 + 13$

Or, $93 - 53 = 40a$

Or, $40 = 40a$

Or, a = 1

Ans: ∴ a = 1

10. If $f(x) = kx + 1$, $g(x) = 3x - 7$ and $gf(1) = 2$, find the value of k.

Solution:

Here,

$f(x) = kx + 1$

$g(x) = 3x - 7$

$gf(1) = 2$

$k = ?$

Now,

$gf(1) = g(k + 1)$

Or, $2 = g(k + 1)$

Or, $2 = 3(k + 1) - 7$

Or, $2 = 3k + 3 - 7$

Or, $2 + 4 = 3k$

Or, $6 = 3k$

Or, $k = 2$

Ans: $\therefore k = 2$

---

11. If $f(x) = 3x - 2$, $fg(x) = 6x - 2$ and $gf(x) = 8$, find the value of x.

Solution:

Here,

$f(x) = 3x - 2$

$fg(x) = 6x - 2$

$gf(x) = 8$

$x = ?$

Now,

$fg(x) = 6x - 2$

Or, $f(g(x)) = 6x - 2$

Or, $3(g(x)) - 2 = 6x - 2$

Or, $3 \cdot g(x) = 6x - 2 + 2$

Or, $3 \cdot g(x) = 6x$

Or, $g(x) = \frac{6x}{3}$

Or, $g(x) = 2x$

Now,

$gf(x) = 8$

Or, $g(3x - 2) = 8$

Or, $2(3x - 2) = 8$

Or, $6x - 4 = 8$

Or, $6x = 8 + 4$

Or, $x = \frac{12}{6}$

Or, $x = 2$

Ans: $\therefore x = 2$